Practical-8

MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Theory

1. Is the schema normalized properly?

Answer: Yes. The database is divided into separate tables like students, events, and registrations. This avoids repeating the same data and keeps the database organized.

2. Are primary keys and foreign keys correctly defined?

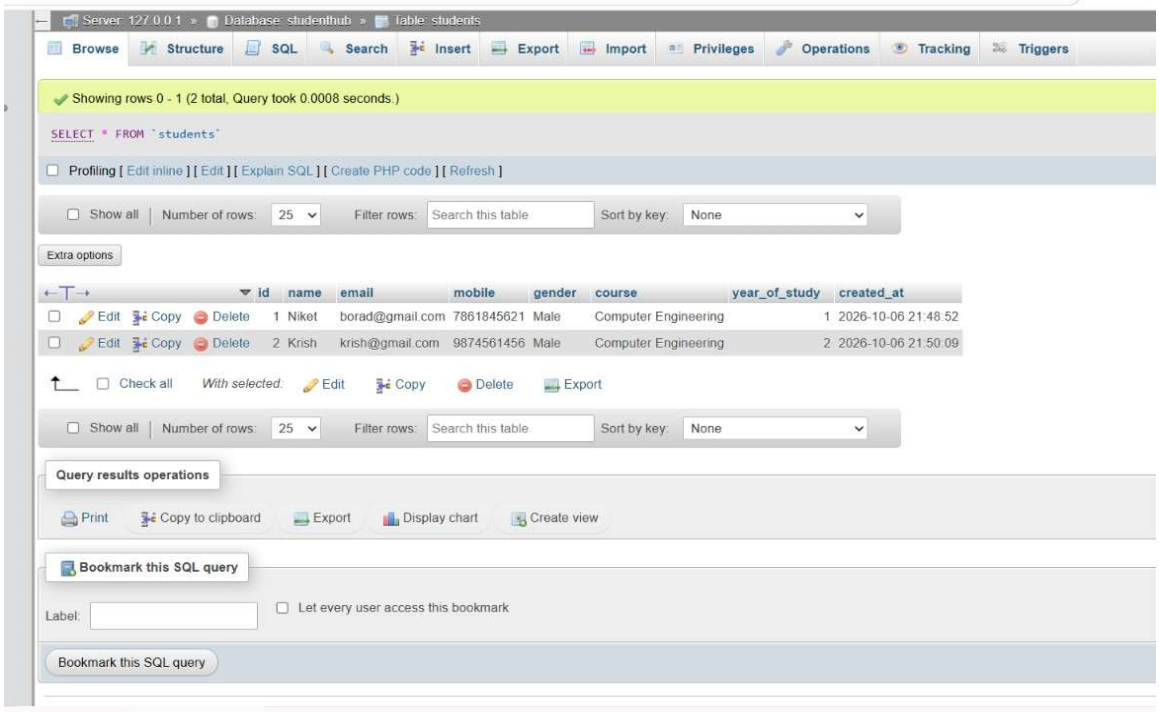
Answer: Yes. Each table has a **primary key** to uniquely identify each record. Foreign keys connect related tables, such as student_id and event_id in the registrations table.

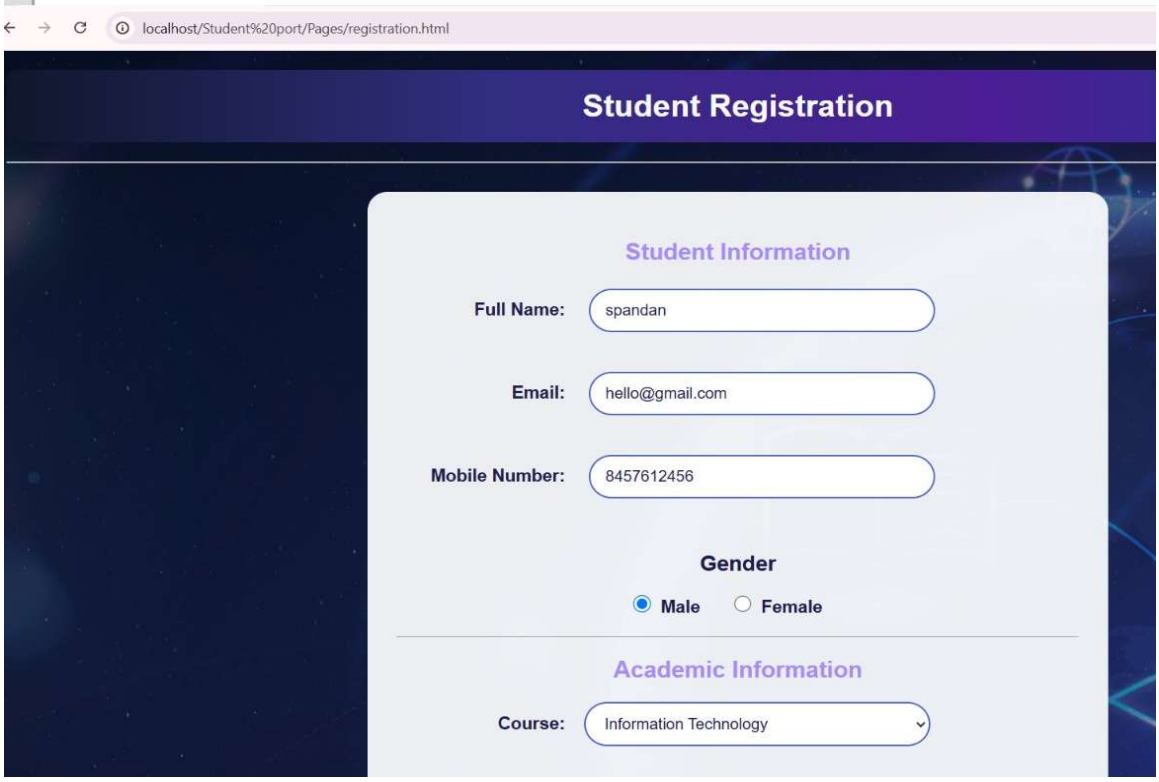
3. Is the database connection handled securely?

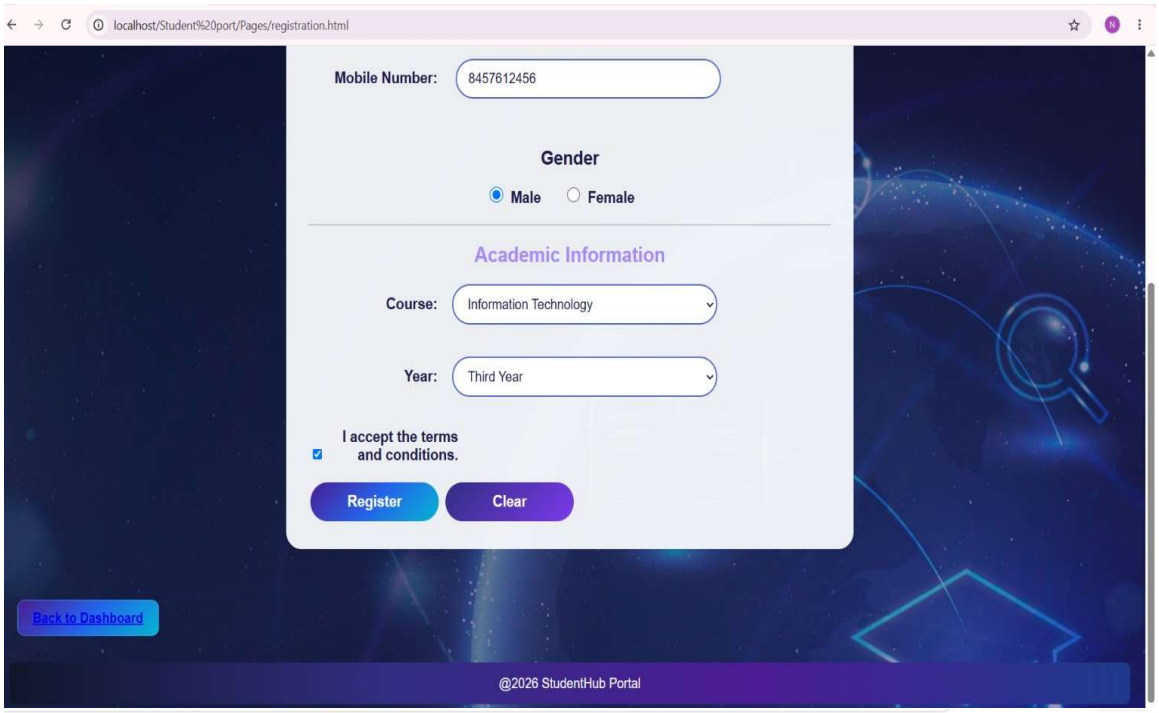
Answer: Yes, reasonably. The connection details are kept in a separate db.php file, and the code uses prepared statements when inserting data.

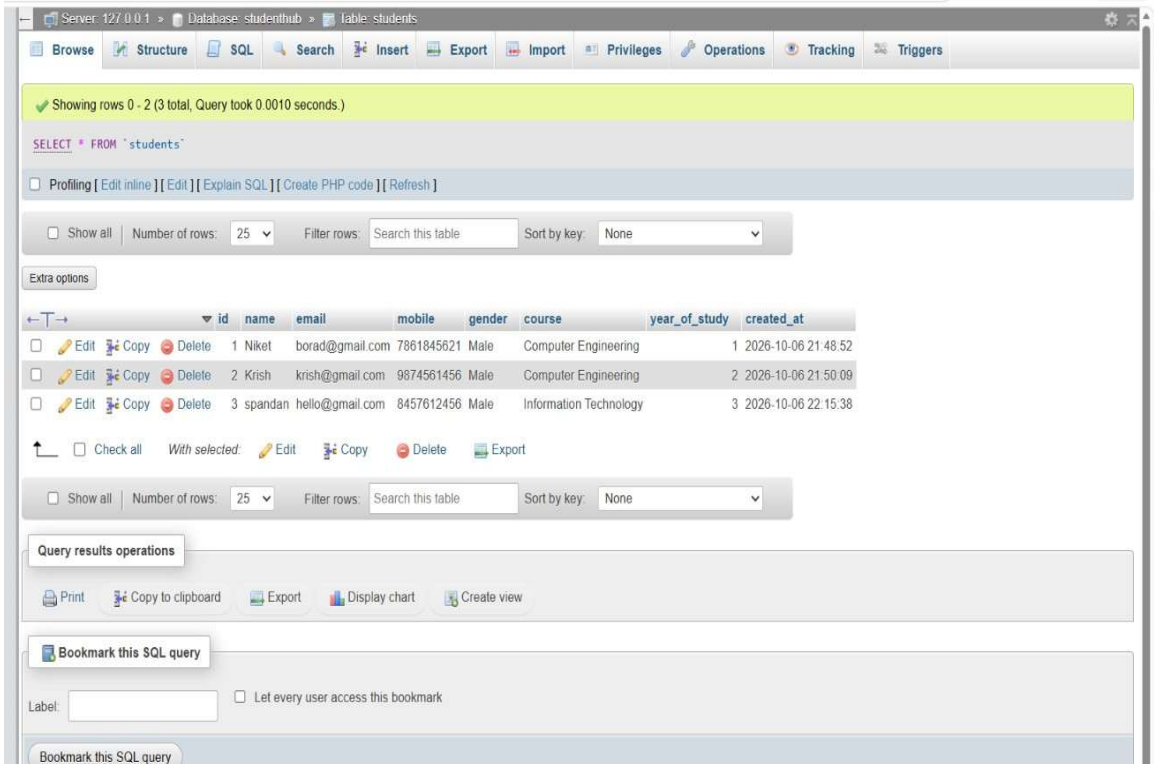
Result:

Evidence of Work (Screenshots)


Server: 127.0.0.1 » Database: studenthub » Table: students
Browse
Structure
SQL
Search
Insert
Export
Import
Privileges
Operations
Tracking
Triggers
Showing rows 0 - 1 (2 total, Query took 0.0008 seconds.)
SELECT * FROM `students`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
id name email mobile gender course year_of_study created_at
Edit Copy Delete 1 Niket borad@gmail.com 7861845621 Male Computer Engineering 1 2026-10-06 21:48:52
Edit Copy Delete 2 Krish krish@gmail.com 9874561456 Male Computer Engineering 2 2026-10-06 21:50:09
Check all With selected: Edit Copy Delete Export
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Query results operations
Print Copy to clipboard Export Display chart Create view
Bookmark this SQL query
Label:
Let every user access this bookmark
Bookmark this SQL query


localhost/Student%20port/Pages/registration.html
Student Registration
Student Information
Full Name: spandan
Email: hello@gmail.com
Mobile Number: 8457612456
Gender
Male
Female
Academic Information
Course: Information Technology





A. GitHub Repository Link:

https://github.com/Niket-Borad/StudentHub-Portal/tree/main/Practical%208

For XAMMP Server Link:-

http://localhost/Student%20port/Pages/registration.html

B. Challenges Faced:

I did not know much about MYSQL at the beginning.
I had difficulty understanding PHP syntax. And How to connect with MYSQL.
I was confused about how to connect my HTML pages with MYSQL.
I did not know how to run my project using the XAMPP server. Also How to Make Table in XAMMP Server also diificult for beginner .
I faced errors while connecting the HTML form with the PHP file. And MYSQL.

C.Learning Outcome:-

Learned to design a normalized MySQL database, connect PHP with MySQL, and use primary/foreign keys and prepared statements securely.

Faculty Remarks: